OKP Seminar

Fixed Raters, Random Selection:
A Correlation-Based Framework for Agreement ICCs in Two-Way Mixed-Effects Designs

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How to measure reliability?

- Reliability asks whether a score can support the same conclusion across replications of a specified measurement procedure.
- Classical Test Theory (Lord et al., 1968)

$$Y_{ij} = T_{ij} + \varepsilon_{ij}, \quad \sigma_Y^2 = \sigma_T^2 + \sigma_\varepsilon^2$$

- The reliability coefficient and **two interpretations**:

$$\rho = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}$$

Correlation Approach

(under the parallel assumptions)

$$\rho = \mathrm{Cor}\left(Y_{ij}, Y_{ij'}\right)$$

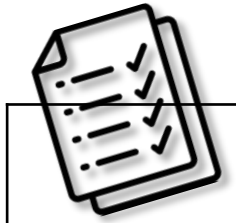
The degree of similarity between two replicates.

Variance Partitioning Approach

$$\rho = \frac{\sigma^2_{\text{person}}}{\sigma^2_{\text{person}} + \sigma^2_{\text{error}}}$$

How well can we distinguish targets from error?

Agreement and consistency



	Rater A	Rater B
S_1	20	30
S_2	35	45
S_3	45	55
S_4	67	77
⋮		
S_{10}	80	90

- Would another rater support the same decision?

1. Do the raters give interchangeable scores?

   ⇒ Interrater **Agreement**

2. Do the raters rank patients similarly?

   ⇒ Interrater **Consistency**

Two-way random-effects designs

	R_1	R_2	R_3	…	R_k
S_1	x	x	x		x
S_2	x	x	x		x
S_3	x	x	x		x
S_4	x	x	x		x
⋮				⋱	
S_n	x	x	x		x

- Person: $i = 1, \ldots, n$, Rater $j = 1, \ldots, k$
- Model:

$$Y_{ij} = \mu + S_i + R_j + \varepsilon_{ij}$$

  - $S_i \sim (0, \sigma_S^2)$; $R_j \sim (0, \sigma_R^2)$; $\varepsilon_{ij} \sim (0, \sigma_\varepsilon^2)$
  - Cannot separate the person×rater interaction from residual error.

- Total variance:

$$\mathrm{Var}(Y_{ij}) = \sigma_S^2 + \sigma_R^2 + \sigma_\varepsilon^2$$

Two-way mixed-effects designs

Fixed raters of interest

	R_1	R_2	R_3	…	R_K
S_1	x	x	x		x
S_2	x	x	x		x
S_3	x	x	x		x
S_4	x	x	x		x
⋮				⋱	
S_n	x	x	x		x

- Person: $i = 1, \ldots, n$, Rater $j = 1, \ldots, K$
- Model:

$$Y_{ij} = \mu + S_i + R_j + \varepsilon_{ij}$$

  - $S_i \sim (0, \sigma_S^2)$; $\varepsilon_{ij} \sim (0, \sigma_\varepsilon^2)$
  - $\sum_{j=1}^{K} R_j = 0$, R_j's are unknown constants.
  - Cannot separate the person×rater interaction from residual error.

- Total variance:

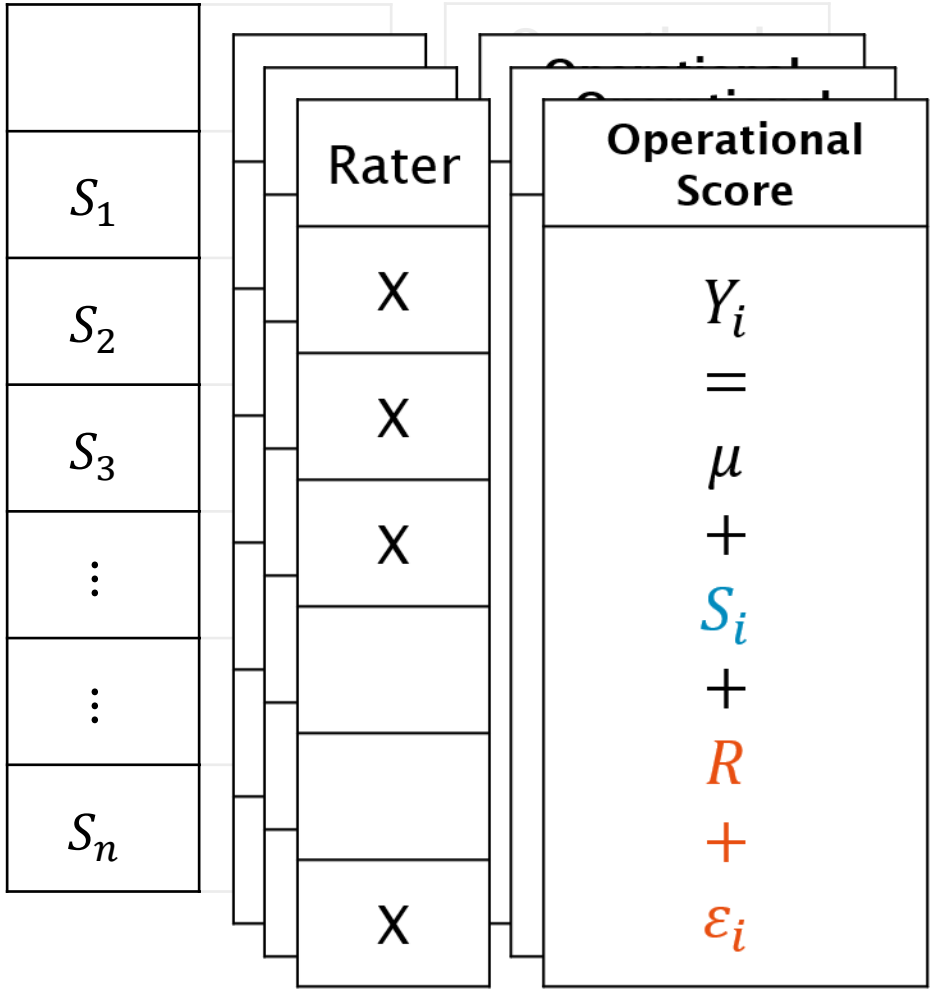
$$\mathrm{Var}(Y_{ij}) = \sigma_S^2 + \sigma_\varepsilon^2$$

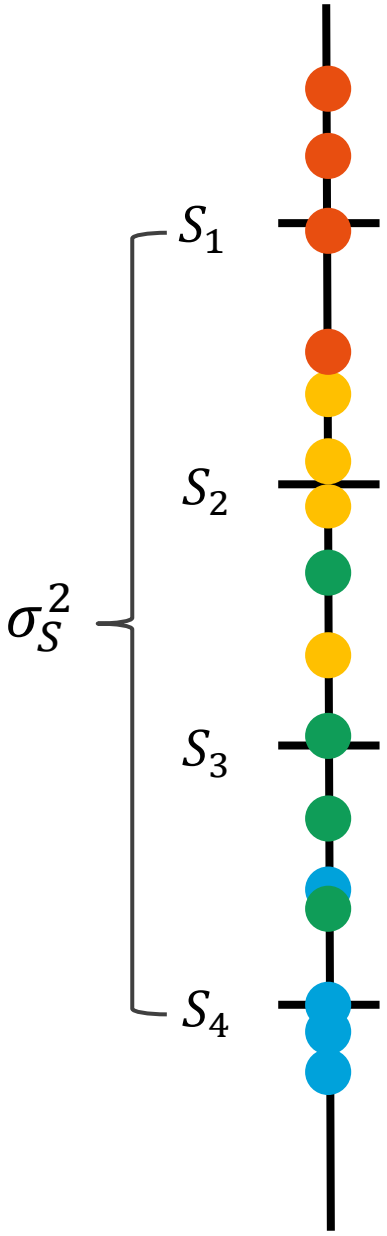
McGraw and Wong's (1996) ICCs

Model	Unit	Decision: Agreement	Decision: Consistency
Two-way random-effects	Single	$\text{ICC(A, 1)} = \frac{\sigma_S^2}{\sigma_S^2+\sigma_R^2+\sigma_\varepsilon^2}$	$\text{ICC(C, 1)} = \frac{\sigma_S^2}{\sigma_S^2+\sigma_\varepsilon^2}$
	Average	$\text{ICC(A, k)} = \frac{\sigma_S^2}{\sigma_S^2+\frac{\sigma_R^2}{k}+\frac{\sigma_\varepsilon^2}{k}}$	$\text{ICC(C, k)} = \frac{\sigma_S^2}{\sigma_S^2+\frac{\sigma_\varepsilon^2}{k}}$
Two-way mixed-effects	Single	$\text{ICC(A, 1)} = \frac{\sigma_S^2}{\sigma_S^2+\theta_R^2+\sigma_\varepsilon^2}$	$\text{ICC(C, 1)} = \frac{\sigma_S^2}{\sigma_S^2+\sigma_\varepsilon^2}$
	Average	$\text{ICC(A, k)} = \frac{\sigma_S^2}{\sigma_S^2+\frac{\theta_R^2}{k}+\frac{\sigma_\varepsilon^2}{k}}$	$\text{ICC(C, k)} = \frac{\sigma_S^2}{\sigma_S^2+\frac{\sigma_\varepsilon^2}{k}}$

$\theta_R^2 = \frac{1}{k-1}\sum R_j^2$: Rater dispersion for fixed rater effects

Variance partitioning interpretation

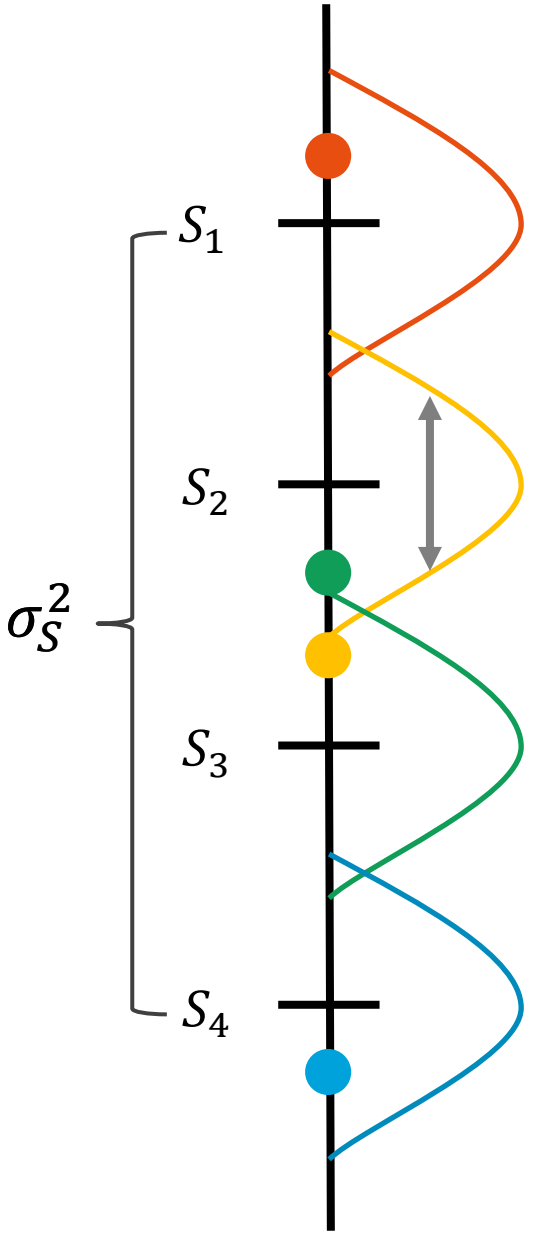




Variance partitioning interpretation

	Rater
S_1	X
S_2	X
S_3	X
⋮	
⋮	
S_n	X

Operational Score
Y_i
$=$
μ
$+$
S_i
$+$
R
$+$
ε_i



$$\rho = \frac{\sigma^2_{\text{person}}}{\sigma^2_{\text{person}} + \sigma^2_{\text{error}}}$$

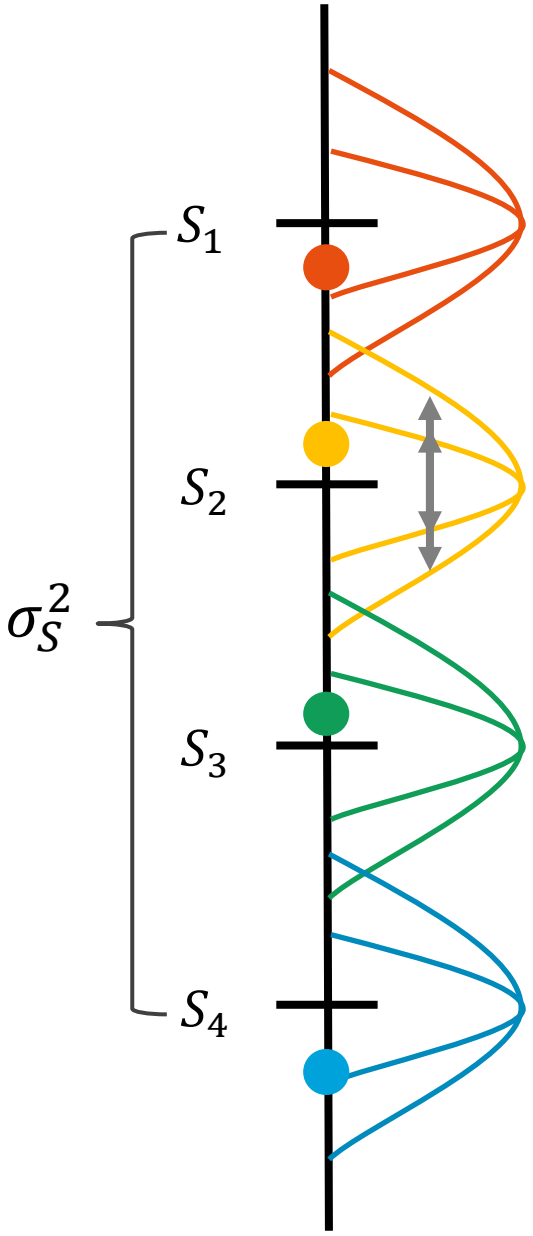
$$\sigma^2_{\text{abs.error}} = \sigma^2_R + \sigma^2_\varepsilon$$

$$\text{ICC}(A,1) = \frac{\sigma^2_s}{\sigma^2_s + \sigma^2_R + \sigma^2_\varepsilon}$$

Single vs. average scores

	R_1	⋯	R_k
S_1	X		X
S_2	X		X
S_3	X		X
⋮			
⋮			
S_n	X		X

Operational Score
$\bar{Y}_{i.}$ = μ + S_i + $\frac{R_1 + \cdots + R_k}{k}$ + $\frac{\varepsilon_1 + \cdots + \varepsilon_k}{k}$



$$\rho = \frac{\sigma^2_{\text{person}}}{\sigma^2_{\text{person}} + \sigma^2_{\text{error}}}$$

$$\sigma^2_{\text{abs.error}} = (\sigma_R^2 + \sigma_\varepsilon^2)/k$$

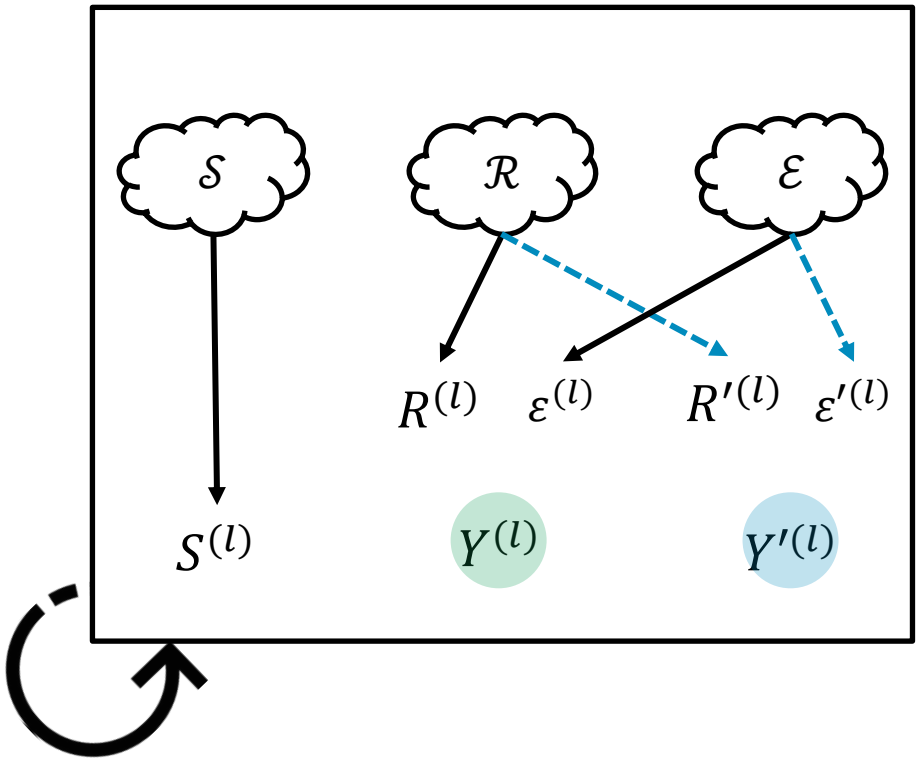
$$\text{ICC}(A,k) = \frac{\sigma_s^2}{\sigma_s^2 + (\sigma_R^2 + \sigma_\varepsilon^2)/k}$$

McGraw and Wong's (1996) ICCs

Model	Unit	Decision: Agreement	Decision: Consistency
Two-way random-effects	Single	ICC(A, 1) $= \mathrm{Cor}(Y_{ij}, Y_{ij'})$	ICC(C, 1) $= \frac{\sigma_S^2}{\sigma_S^2+\sigma_\varepsilon^2}$
	Average	ICC(A, k) $= \mathrm{Cor}(\bar{Y}_{i.}, \bar{Y}_{i.'})$	ICC(C, k) $= \frac{\sigma_S^2}{\sigma_S^2+\frac{\sigma_\varepsilon^2}{k}}$
Two-way mixed-effects	Single	ICC(A, 1) $= \frac{\sigma_S^2}{\sigma_S^2+\theta_R^2+\sigma_\varepsilon^2}$	ICC(C, 1) $= \frac{\sigma_S^2}{\sigma_S^2+\sigma_\varepsilon^2}$
	Average	ICC(A, k) $= \frac{\sigma_S^2}{\sigma_S^2+\frac{\theta_R^2}{k}+\frac{\sigma_\varepsilon^2}{k}}$	ICC(C, k) $= \frac{\sigma_S^2}{\sigma_S^2+\frac{\sigma_\varepsilon^2}{k}}$

$\theta_R^2 = \frac{1}{k-1}\sum R_j^2$: Rater dispersion for fixed rater effects

Agreement ICC (Shrout & Fleiss, 1979)



Hypothetical replicate

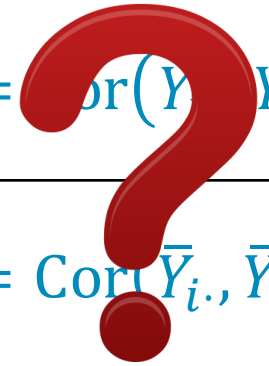
	$R^{(1)}$	$R^{(2)}$	$R^{(3)}$	...
$S^{(1)}$	$Y^{(1)}$			
$S^{(2)}$		$Y^{(2)}$		
$S^{(3)}$			$Y^{(3)}$	
⋮				⋱

	$R'^{(1)}$	$R'^{(2)}$	$R'^{(3)}$	...
$S^{(1)}$	$Y'^{(1)}$			
$S^{(2)}$		$Y'^{(2)}$		
$S^{(3)}$			$Y'^{(3)}$	
⋮				⋱

$$\mathrm{Cor}(Y_{ij}, Y_{ij'}) = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_R^2 + \sigma_\varepsilon^2} = \mathrm{ICC(A,1)}$$

McGraw and Wong's (1996) ICCs

Model	Unit	Decision: Agreement	Decision: Consistency
Two-way random-effects	Single	ICC(A, 1) $= \mathrm{Cor}(Y_{ij}, Y_{ij'})$	ICC(C, 1) $= \frac{\sigma_s^2}{\sigma_s^2+\sigma_\varepsilon^2}$
	Average	ICC(A, k) $= \mathrm{Cor}(\bar{Y}_{i.}, \bar{Y}_{i.'})$	ICC(C, k) $= \frac{\sigma_s^2}{\sigma_s^2+\frac{\sigma_\varepsilon^2}{k}}$
Two-way mixed-effects	Single	ICC(A, 1) $= \mathrm{Cor}(Y_{ij}, Y_{ij'})$	ICC(C, 1) $= \frac{\sigma_s^2}{\sigma_s^2+\sigma_\varepsilon^2}$
	Average	ICC(A, k) $= \mathrm{Cor}(\bar{Y}_{i.}, \bar{Y}_{i.'})$	ICC(C, k) $= \frac{\sigma_s^2}{\sigma_s^2+\frac{\sigma_\varepsilon^2}{k}}$



$\theta_R^2 = \frac{1}{k-1}\sum R_j^2$: Rater dispersion for fixed rater effects

Fixed-rater agreement paradox (1)

- Rater effects $R_1, \ldots, R_K$ are the fixed, unknown constants, $\sum_{j=1}^{K} R_j = 0$.
  - There is no probability variance,

$$\mathrm{Cor}\left(Y_{ij}, Y_{ij'}\right) = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_\varepsilon^2} = \mathrm{ICC}(C, 1) \qquad \text{when } j \neq j'$$

$$\nparallel$$

$$\mathrm{ICC}(\mathrm{A}, 1) = \frac{\sigma_s^2}{\sigma_s^2 + \theta_R^2 + \sigma_\varepsilon^2}$$

(Recall that $\mathrm{ICC}(A, 1) = \mathrm{Cor}\left(Y_{ij}, Y_{ij'}\right)$ in two-way random-effects models.)

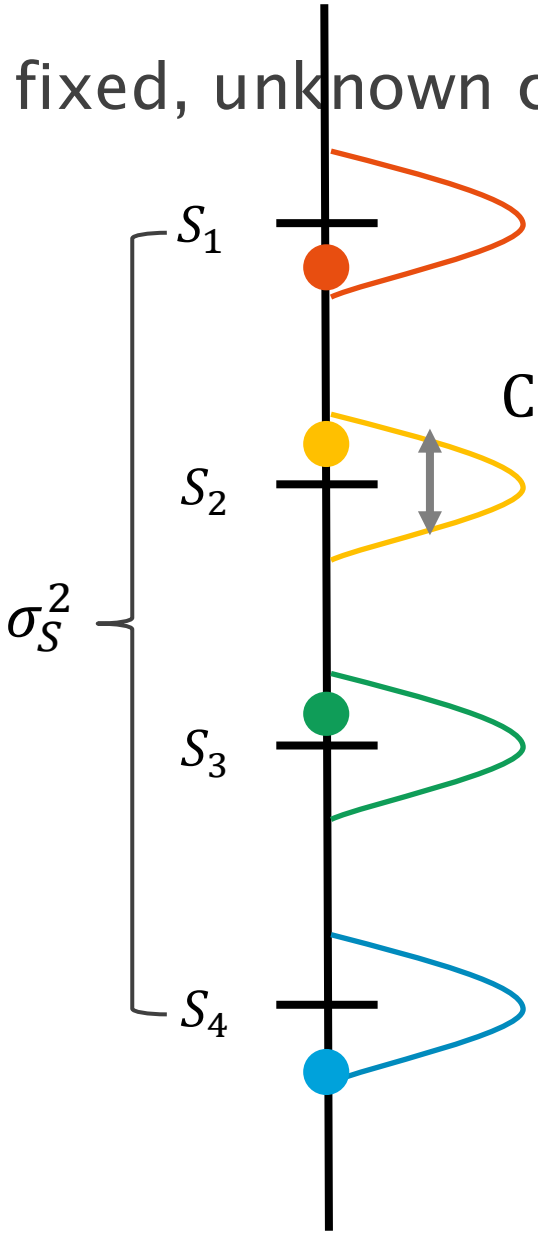
Fixed-rater agreement paradox (2)

- Rater effects $R_1, \ldots, R_K$ are the fixed, unknown constants, $\sum_{j=1}^{K} R_j = 0$.

	R_1	…	R_K
S_1	x		x
S_2	x		x
S_3	x		x
S_4	x		x
⋮		⋱	
S_n	x		x

Operational Score
$\bar{Y}_{i\cdot} = \mu + S_i + \frac{R_1 + \cdots + R_K}{K} + \frac{\varepsilon_1 + \cdots + \varepsilon_K}{K}$

$\frac{R_1 + \cdots + R_K}{K} = 0$



$$\mathrm{Cor}(\bar{Y}_{i\cdot}, \bar{Y}'_{i\cdot}) = \frac{\sigma_s^2}{\sigma_s^2 + \frac{\sigma_\varepsilon^2}{K}} = \mathrm{ICC}(C, K)$$

$\neq$

$$\mathrm{ICC}(\mathrm{A}, K) = \frac{\sigma_s^2}{\sigma_s^2 + \frac{\theta_R^2}{K} + \frac{\sigma_\varepsilon^2}{K}}$$

$\sigma^2_{\text{abs.error}}$?

Fixed-rater dispersion still matters

- Dispersion among fixed raters θ_R^2
  - Fixed raters can differ systematically in severity.
  - Under a variance-partitioning interpretation, the dispersion θ_R^2 is included in the denominator as **a source of disagreement.**

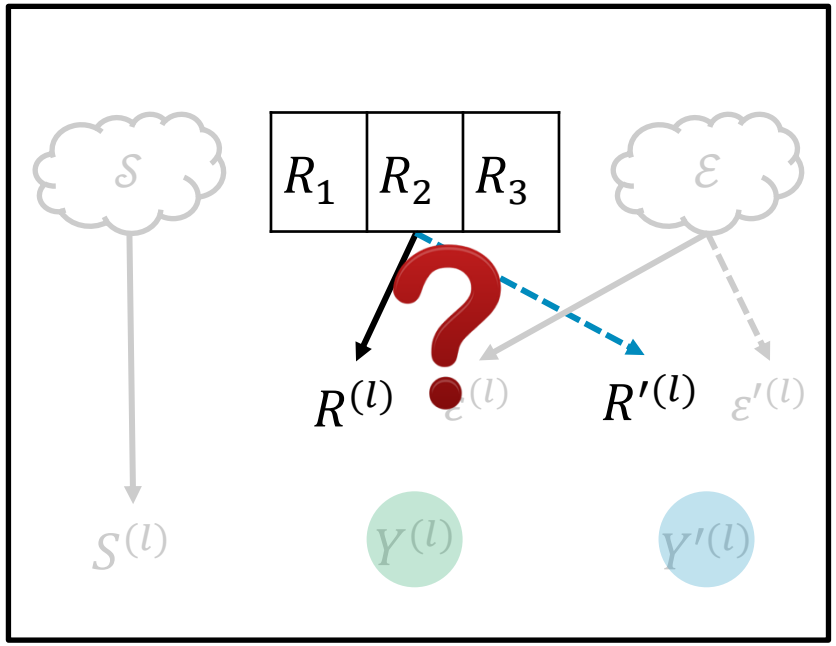
How can these rater differences contribute to a correlation between replicate scores?

- Fixed rater, random selection
  - J: a random index selected uniformly from $\{1, \ldots, K\}$
  - R_J: a random variable / R_j: a fixed rater effect
  - $\delta_R^2 = \mathrm{Var}(R_J) = \frac{1}{K}\sum_{j=1}^{K} R_j^2$: finite-population variance

(vs. $\theta_R^2 = \frac{1}{K-1}\sum_{j=1}^{K} R_j^2$: ANOVA-scaled dispersion)

Agreement ICC for single scores (1)

- **Single score $k = 1, K = 3$**



	R_1	R_2	R_3
$S^{(1)}$	x		
$S^{(2)}$		x	
$S^{(3)}$			x
$S^{(4)}$	x		
⋮			

Hypothetical replicate

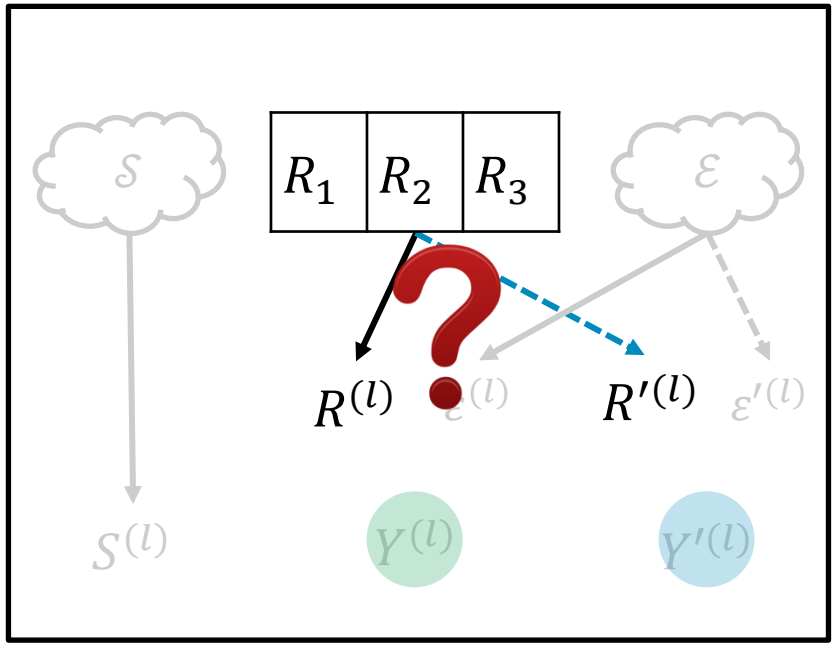
	R_1	R_2	R_3
$S^{(1)}$			x
$S^{(2)}$	x		
$S^{(3)}$		x	
$S^{(4)}$		x	
⋮			

Between: Sampling without replacement
i.e., $J_1 \neq J_2$ or overlap $M = 0$

$$\mathrm{ICC}^*(\mathrm{A}, k = 1 \mid M = 0) = \mathrm{Cor}(Y_{iJ_1}, Y_{iJ_2} \mid J_1 \neq J_2) = \frac{\sigma_s^2 - \delta_R^2/2}{\sigma_s^2 + \delta_R^2 + \sigma_\varepsilon^2}$$

Agreement ICC for single scores (2)

- **Single score $k = 1, K = 3$**



	R_1	R_2	R_3
$S^{(1)}$	x		
$S^{(2)}$		x	
$S^{(3)}$			x
$S^{(4)}$	x		
⋮			

Hypothetical replicate

	R_1	R_2	R_3
$S^{(1)}$		x	
$S^{(2)}$		x	
$S^{(3)}$	x		
$S^{(4)}$	x		
⋮			

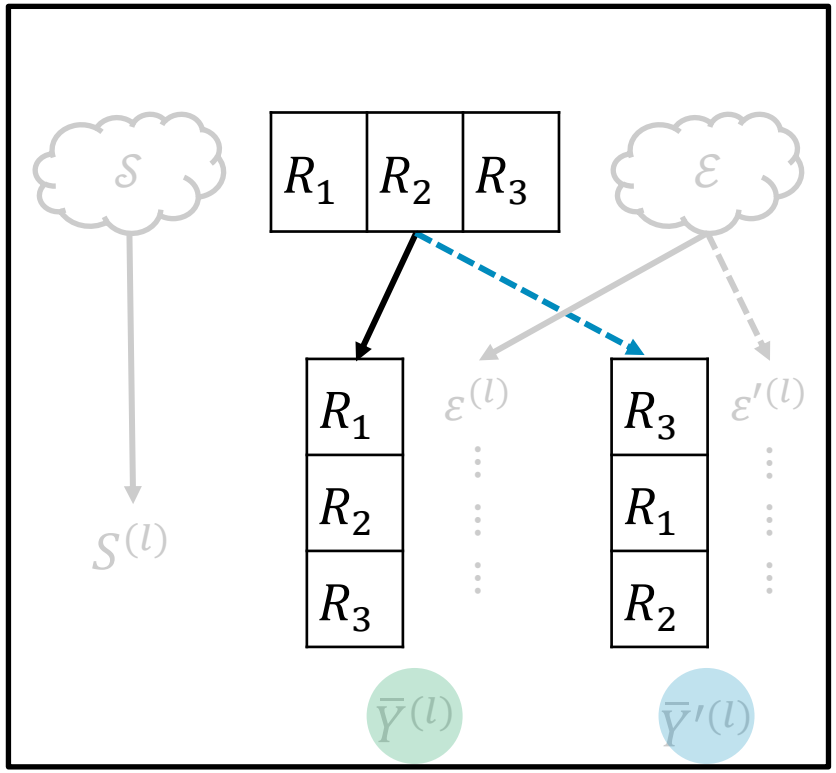
Between: Sampling with replacement
i.e., M could be 0 or 1

vs. $\mathrm{ICC}(\mathrm{A},1) = \dfrac{\sigma_s^2}{\sigma_s^2 + \theta_R^2 + \sigma_\varepsilon^2}$

$$\mathrm{ICC}^*(A, k=1) = \mathrm{Cor}(Y_{iJ_1}, Y_{iJ_2}) = \frac{\sigma_s^2}{\sigma_s^2 + \delta_R^2 + \sigma_\varepsilon^2}$$

Agreement ICC for all-rater average scores

- **Average score $k = K = 3$**



	R_1	R_2	R_3
$S^{(1)}$	x	x	x
$S^{(2)}$	x	x	x
$S^{(3)}$	x	x	x
$S^{(4)}$	x	x	x
⋮			

Hypothetical replicate

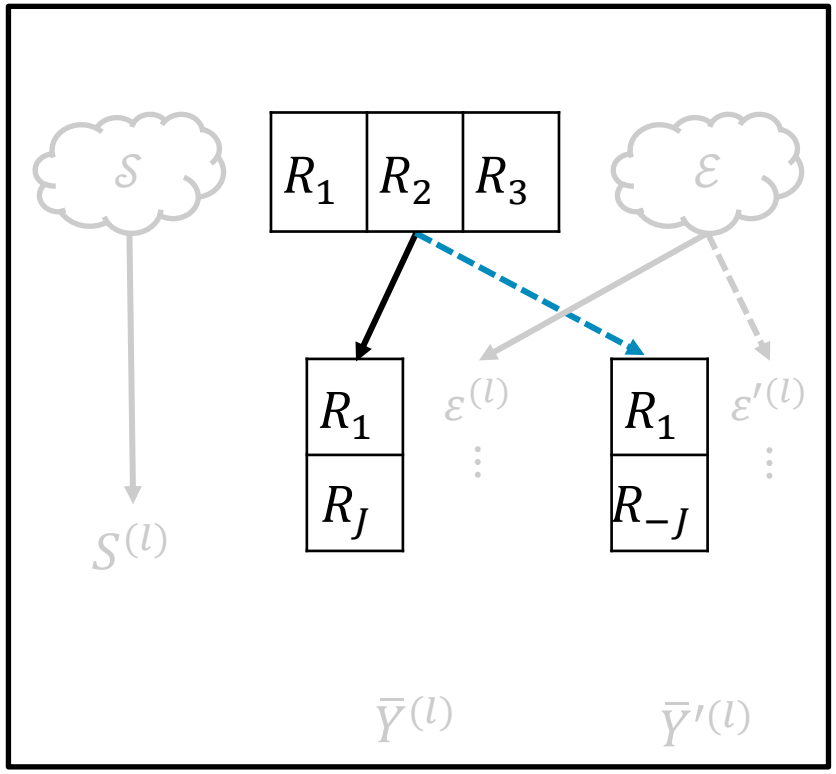
	R_1	R_2	R_3
$S^{(1)}$	x	x	x
$S^{(2)}$	x	x	x
$S^{(3)}$	x	x	x
$S^{(4)}$	x	x	x
⋮			

Within: Sampling without replacement
Between: Sampling MUST with replacement. i.e., $M = 3$

$$\mathrm{ICC}^*(A, k = 3) = \mathrm{ICC}^*(A, k = 3 \mid M = 3) = \mathrm{Cor}(\bar{Y}_{i\cdot}, \bar{Y}'_{i\cdot}) = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_\varepsilon^2/3}$$

Agreement ICC for partial-rater average scores

- **Average score $k = 2, K = 3$**



	R_1	R_2	R_3
$S^{(1)}$	x	x	
$S^{(2)}$	x	x	
$S^{(3)}$	x		x
$S^{(4)}$	x		x
⋮			

Hypothetical replicate

	R_1	R_2	R_3
$S^{(1)}$	x		x
$S^{(2)}$	x		x
$S^{(3)}$	x	x	
$S^{(4)}$	x	x	
⋮			

Within: Sampling without replacement
Between: Sampling with partial replacement. i.e., $M = 1$

$$\mathrm{ICC}^*(A, k = 2 \mid M = 1) = \frac{\sigma_s^2 - \delta_R^2/8}{\sigma_s^2 + \delta_R^2/4 + \sigma_\varepsilon^2/2}$$

The family of agreement ICCs

Distinct rater reliability

Under specific planning-missing designs

Common settings

$\bar{Y}_{i,\mathcal{J}_1} \leftrightarrow \bar{Y}'_{i,\mathcal{J}_2}$

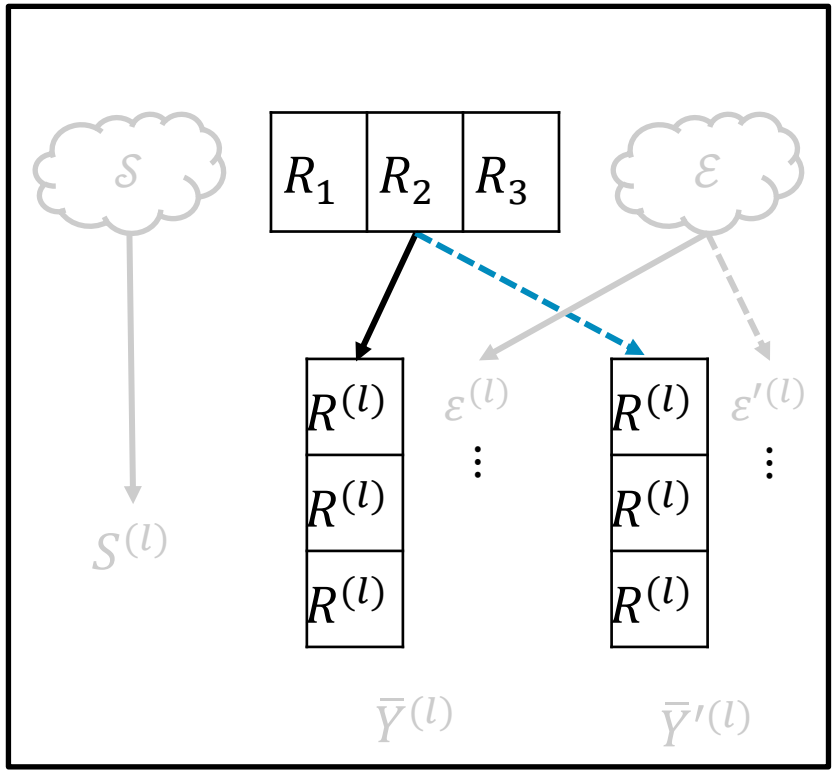
replicates

Within replicates	Without replacement (Non-overlap, $m = 0$)	With partial replacement (Partial overlap, $1 \leq m \leq k$)	With replacement (Random overlap, M is RV)
Without replacement (k distinct rater)	$\mathrm{ICC}^*(A,k \mid M=0) = \frac{\sigma_S^2 - \frac{\delta_R^2}{K-1}}{\sigma_S^2 + \frac{\delta_R^2}{k}\left(\frac{K-k}{K-1}\right) + \frac{\sigma_\varepsilon^2}{k}}$	$\mathrm{ICC}^*(A,k \mid M=m) = \frac{\sigma_S^2 + \frac{\delta_R^2}{k^2}\left(\frac{mK-k^2}{K-1}\right)}{\sigma_S^2 + \frac{\delta_R^2}{k}\left(\frac{K-k}{K-1}\right) + \frac{\sigma_\varepsilon^2}{k}}$	$\mathrm{ICC}^*(A,k) = \frac{\sigma_S^2}{\sigma_S^2 + \frac{\delta_R^2}{k}\left(\frac{K-k}{K-1}\right) + \frac{\sigma_\varepsilon^2}{k}}$
With replacement (k iid sampling)	—	—	$\mathrm{ICC}^*_{\mathrm{iid}}(A,k) = \frac{\sigma_S^2}{\sigma_S^2 + \frac{\delta_R^2}{k} + \frac{\sigma_\varepsilon^2}{k}}$

Structurally identical to conventional ICC(A, k)

Link to conventional agreement ICC

- **Average score $k = K = 3$**



	R_1	R_2	R_3
$S^{(1)}$	xx	x	
$S^{(2)}$		xxx	
$S^{(3)}$		x	xx
$S^{(4)}$	xx		x
⋮			

Hypothetical replicate

	R_1	R_2	R_3
$S^{(1)}$	x	x	x
$S^{(2)}$	x		xx
$S^{(3)}$	xx	x	
$S^{(4)}$		x	xx
⋮			

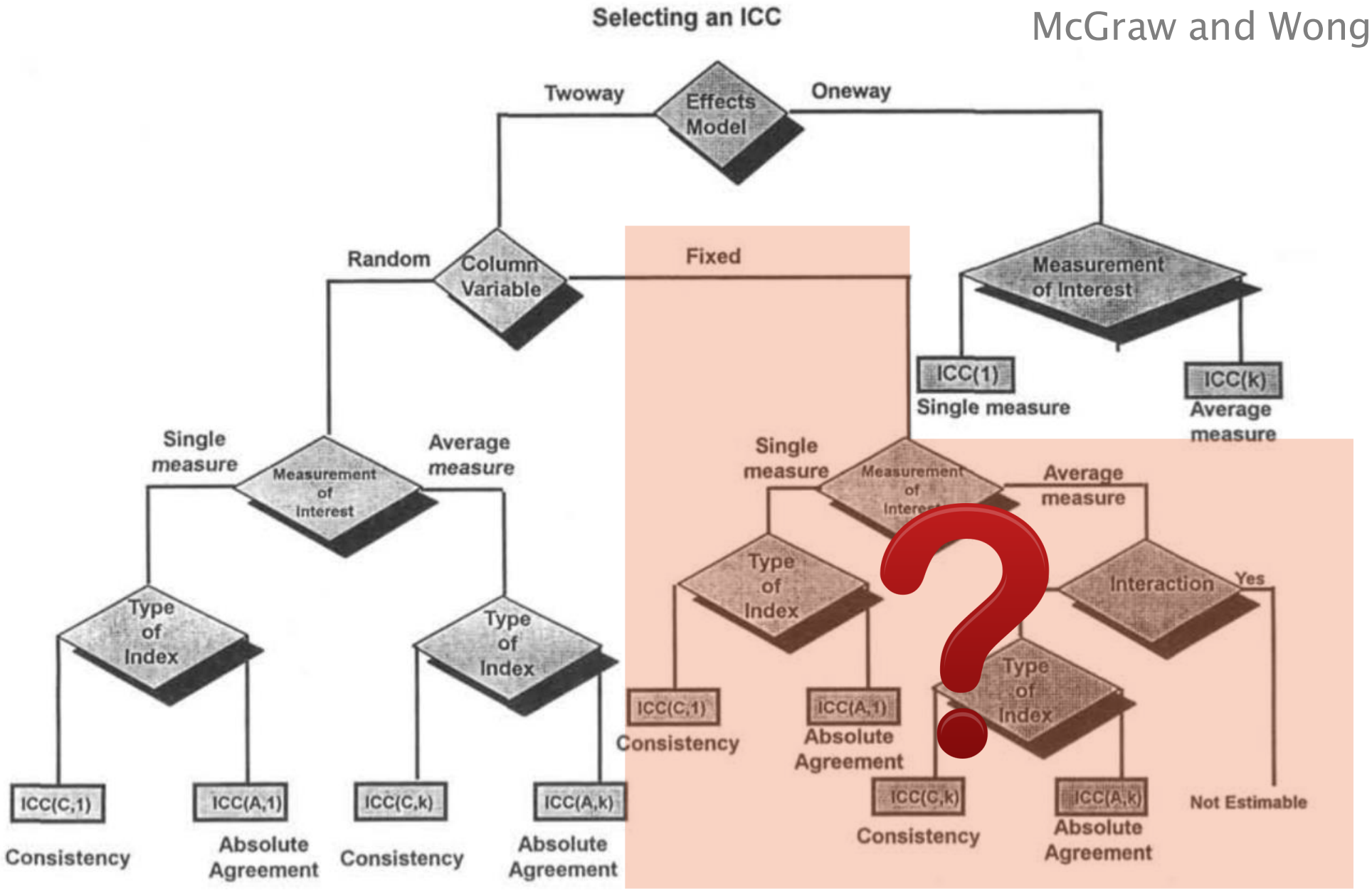
Within: Sampling with replacement
Between: Sampling with replacement. i.e., M is a RV

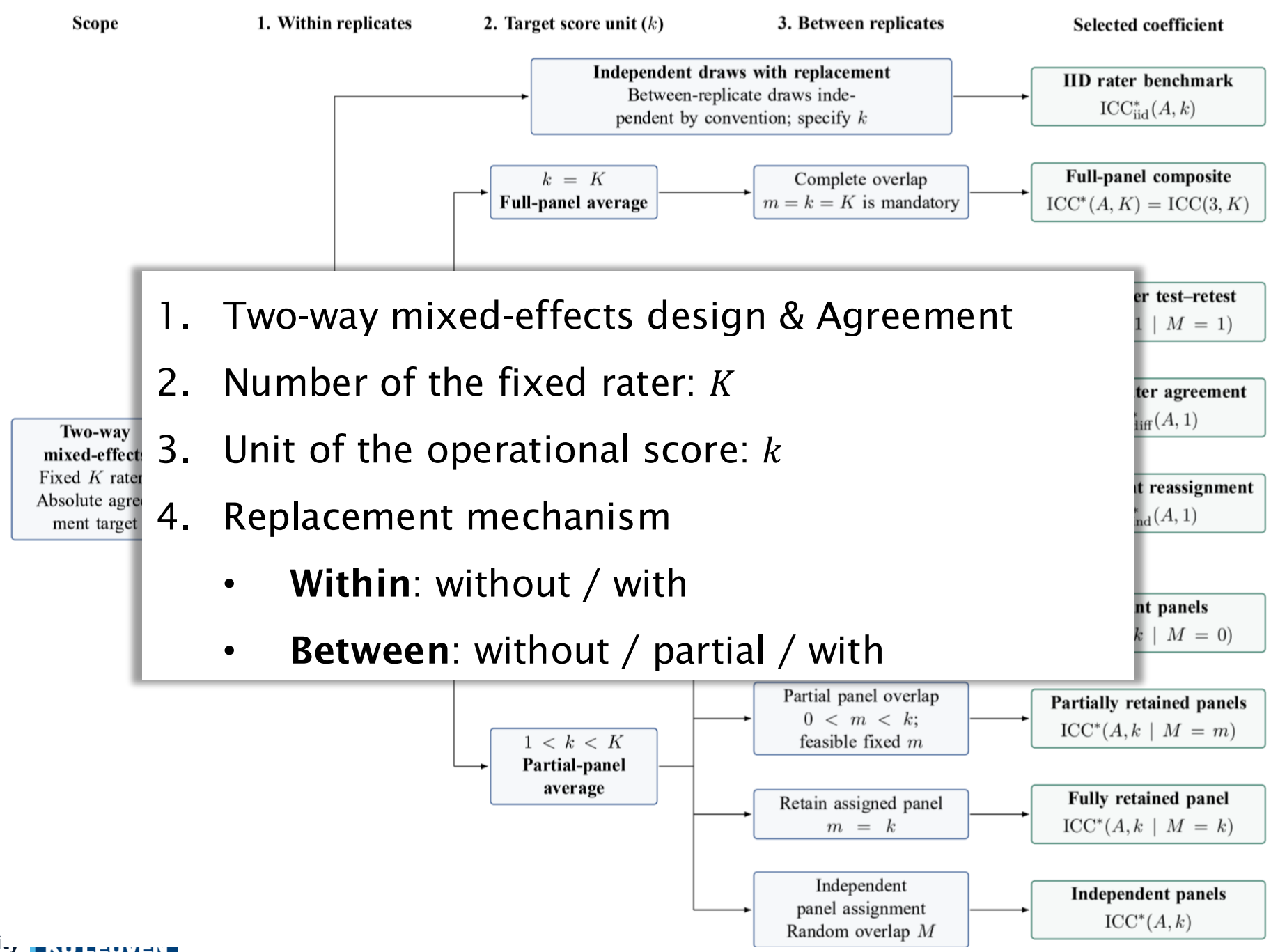
$$\mathrm{ICC}^{*}_{\mathrm{iid}}(A,\mathrm{K}) = \frac{\sigma_S^2}{\sigma_S^2 + \frac{\delta_R^2}{K} + \frac{\sigma_\varepsilon^2}{K}}$$

vs. $$\mathrm{ICC}(\mathrm{A},\mathrm{K}) = \frac{\sigma_s^2}{\sigma_s^2 + \frac{\theta_R^2}{K} + \frac{\sigma_\varepsilon^2}{K}}$$

Flowchart for selecting an ICC

McGraw and Wong (1996)





1. Two-way mixed-effects design & Agreement
2. Number of the fixed rater: K
3. Unit of the operational score: k
4. Replacement mechanism
   - **Within**: without / with
   - **Between**: without / partial / with

Ex 1: Single ratings from 3 fixed raters

- **Math achievement measures.**

  $n = 161$ (Delacruz, 2011. pp. 33, 127)

  The three open-ended items on the math assessment were scored using three raters who were trained on how to apply the rubrics…….
  Each rater used the final version of the rubric to score responses from **20 students**.
  Interrater reliability for each item was determined by calculating ICC for each item, which were applied to single measurement scores (i.e., individual ratings of items by individual judges)……
  A two-way, mixed-effect model was used to examine the absolute agreement of measurements between the three raters for each of the three open-ended items.

1. Care about interrater agreement
2. Fixed raters, K=3
3. Single rating, k = 1
4. Replication mechanism?

Different raters for two replicates

$$\mathrm{ICC}^*(A, 1 \mid M = 0)$$

Independent reassignment

$$\mathrm{ICC}^*(A, 1)$$

Discussion

- Main contributions
  - Correlation-based definition for ICC
  - Fixed effects and random assignment
  - A family of agreement ICCs for two-way mixed-effects designs
  - To select a proper agreement ICC, researchers should specify the unit of the score, within- and between-replicates sampling mechanism.

- Limitations and Future Work
  - Depend on the model assumptions
  - Estimation properties
  - Practical selection and reporting guidelines
  - Extensions beyond uniform assignment